

Mayushan Mayurathan

(647) 297-8115 | mayushan.mayurathan@mail.utoronto.ca | [linkedin.com/in/mayushan/](https://www.linkedin.com/in/mayushan/) | github.com/mayushanmayurathan

EDUCATION

University of Toronto

Sep 2021 – June 2026

Honours Bachelor of Science in Economics & Computer Science

- **Relevant Computer Coursework:** Database Management Systems, Data Structures & Algorithms, Operating Systems, Systems Programming, Discrete Mathematics
- **Relevant Economics/Statistics Coursework:** Capital Markets & Asset Pricing, Corporate Finance, Exploratory Data Analysis, Applied Econometrics, Algorithms of Data Analysis

EXPERIENCE

Software Systems Analyst

Sept 2024 - Dec 2024

Royal Bank of Canada (RBC)

Toronto, ON

- Engineered **data extraction pipelines** by integrating **SQL queries** with **Python** using the **pyodbc** library
- **Automated CPU Utilization Reporting** through a combination of **Ansible** and **python scripting**
- Administered Unix scripts for Oracle database maintenance to minimize discrepancies across server environments
- Maintained over **300+ Oracle and Sybase system devices** by regulating **ServiceNow** compliance

Quantitative Analyst

May 2024 – Aug 2024

Northstar Trading

Toronto, ON

- Developed a high-performance predictive model for power markets (**88% R^2**), applying **regression and machine learning**; **outperformed existing benchmarks by 4.5% in forecast accuracy**
- Built **low-latency Python ETL pipelines** to transform and stream data to traders giving access to actionable signals
- Designed **statistical arbitrage** strategies for congestion spreads using **non-parametric tests** and **time-series analysis**
- Modeled ISO grid topology as a graph structure using PandaPower and MATLAB; explored graph neural networks in TensorFlow to capture network-level dynamics and forecast disruptions

Mobile App Developer

Jan 2024 – May 2024

Fibra Inc.

Toronto, ON

- Developed Bluetooth support and data storage app features using **React** and **JavaScript** for iOS/Android compatibility
- Engineered and designed Fibra Inc's **Firestore database** to improve data quality and **cut redundancy by 80%**
- Engineered seamless and reliable **Wifi and Bluetooth data paths** achieving **99% uptime**

Quantitative Analyst

January 2024 – March 2024

Antarctic Institute of Canada

Edmonton, AB (Remote)

- Refactored 3 extensive datasets using **SQL** and **Excel**, identifying and addressing data gaps through meticulous research
- Constructed HTML web-scrappers using **BeautifulSoup & Pandas (Python)** to extract 350+ observations
- Communicated findings to supervisors through written reports, presentations and visualizations

PROJECTS

Research Paper: Economics of the Housing Market | [Python](#), [NumPy](#), [Pandas](#), [GeoPandas](#), [scikit-learn](#), [SQL](#)

- Authored a 70+ page economic paper analyzing the effects real estate investing on housing affordability and social welfare
- Managed 1.4 million+ observations through **Python & SQL** for analysis & **Matplotlib & GeoPandas** for visualization
- Developed a detailed presentation of the findings to a distinguished panel, including a professor and 3+ PhD candidates

Transparent Credit Risk Modeling | [Python](#), [R](#), [scikit-learn](#), [Keras/TensorFlow](#), [GLMNet](#), [NumPy](#), [Pandas](#)

- Built credit risk models with Ridge Regression & neural networks (AUC 0.98, 99% accuracy) on 60+ financial variables
- Enhanced Altman Z-score with AIC feature selection and GLMNet, raising specificity from 7% to 78% on distressed firms
- Audited CRA bias across 2,400+ firms, identifying inflated ratings and validating early-warning signals for bankruptcies.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML/CSS, C/C++, R, VBA, MIPS Assembly, Bash, LaTeX

Technologies: React, AWS, Axios, Firebase, Postman, JSON, STATA, JavaFX, ControlsFX, JUnit, Cohere

Developer Tools: Git, Figma, Expo, Miro, Linux, Jupyter, Collab, Power BI

Libraries: LangChain, Pandas, GeoPandas, NumPy, SciPy, Matplotlib, Beautiful Soup, OpenPyXL, DeepFace, OpenCV